HBL Engineering Limited

Rail Signaling Group

In-Process QC Checklist for Stationary Kavach Installation

QC Engineer Name: sushma v1.2

Station ID	Station Name	Zone	Division	Section Name	Initial Date	Updated Date
576765	yg ghh	ECR	Pt Deendayal Upadhy - Pradhankhnta	Pt Deendayal Upadhy - Pradhankhnta	24/03/2026	24/03/2026

Total Points	Closed Points	Open Points
151	166	-15

Installation Status	☐ Completed	☒ Not Completed

1.0 Serial Number Verification

S.No	Units	Status	Observation	Image
1.1	Stationary Kavach Unit - 5666666666664	Matching		
1.2	Peripheral Processing Card 1 - 5666666666664	Matching		
1.3	Peripheral Processing Card 2 - 78888888888888	Matching		
1.4	Vital Computer Card -1 - 5444444977	Matching		
1.5	Vital Computer Card -2	Not Applicable		
1.6	Vital Computer Card -3	Not Applicable		
1.7	Voter Card -1	Not Applicable		
1.8	Voter Card -2	Not Applicable		
1.9	Vital Gateway Card 1 (S2S)	Matching		
1.10	Vital Gateway Card 2 (S2S)	Matching		
1.11	Vital Gateway Card 3 (NMS)	Matching		
1.12	EI Gateway-1	Matching		
1.13	EI Gateway-2	Matching		
1.14	Field Scanner Card 1	Matching		
1.15	Field Scanner Card 2	Matching		
1.16	Field Scanner Card 3	Matching		
1.17	Field Scanner Card 4	Matching		
1.18	Field Scanner Card 5	Matching		

S.No	Units	Status	Observation	Image
1.19	Field Scanner Card 6	Matching		
1.20	Field Scanner Card 7	Matching		
1.21	Field Scanner Card 8	Matching		
1.22	RIU communication card 1-Host	Matching		
1.23	RIU communication card 2-Host	Matching		
1.24	RS 232-OFC converter 1 (STATION)	Matching		
1.25	RS 232-OFC converter 2 (STATION)	Matching		
1.26	RS 485-OFC converter (STATION)	Matching		
1.27	FIU Termination Card 1	Matching		
1.28	FIU Termination Card 2	Matching		
1.29	FIU Termination Card 3	Matching		
1.30	FIU Termination Card 4	Matching		
1.31	FIU Termination Card 5	Matching		
1.32	FIU Termination Card 6	Matching		
1.33	FIU Termination Card 7	Matching		
1.34	FIU Termination Card 8	Matching		
1.35	DPS Card 1	Matching		
1.36	DPS Card 2	Matching		
1.37	GPS & GSM Antenna-1	Matching		
1.38	GPS & GSM Antenna-2	Matching		
1.39	SMOCIP Unit	Matching		
1.40	SMOCIP Termination Panel	Matching		
1.41	Station Termination Panel	Not Applicable		
1.42	Stationary PDU Box	Not Applicable		
1.43	IPS PDU Box	Not Applicable		
1.44	DC-DC Converter (Relay racks)	Not Applicable		
1.45	RTU-1	Not Applicable		
1.46	RTU-2	Not Applicable		
1.47	Station Radio Power Supply card-1	Matching		
1.48	Station Radio Power Supply card-2	Matching		
1.49	Next Gen/. Cal Amp Radio Modem-1	Matching		
1.50	Next Gen/. Cal Amp Radio Modem-2	Matching		

S.No	Units	Status	Observation	Image
1.51	RS 232-OFC converter 1 (RTU-1)	Matching		
1.52	RS 232-OFC converter 2 (RTU-2)	Matching		
1.53	RS 485-OFC converter (SM-OCIP)	Matching		
1.54	RIU	Matching		
1.55	RIU Power Supply Card-1	Matching		
1.56	RIU Power Supply Card-2	Matching		
1.57	RIU communication card 1-Remote	Matching		
1.58	RIU communication card 2-Remote	Matching		
1.59	Field Scanner Card 1	Matching		
1.60	Field Scanner Card 2	Matching		
1.61	Field Scanner Card 3	Matching		
1.62	Field Scanner Card 4	Matching		
1.63	RIU Battery Charge Cum Filter-1	Matching		
1.64	RIU Battery Charge Cum Filter-2	Matching		

2.0 Building (New Building Constructed by HBL)	**Not Applicable**

3.0 Tower (New Tower Constructed by HBL)	**Not Applicable**

4.0 Power Supply

S.No	Aspect	Requirement	Status	Observation	Image
4.1 PDU(Power Distribution Unit)					
4.1.1	Installation	IPS PDU unit shall be mounted on the wall using M8 insulators, secured with M8 bolts, and tightened to a torque of 20 Nm as per diagram 5 16 76 0053	Yes		
4.1.2	Installation	Station PDU unit shall be mounted on the wall using M8 insulators, secured with M8 bolts, and tightened to a torque of 20 Nm as per diagram 5 16 76 0054	Yes		

S.No	Aspect	Requirement	Status	Observation	Image
4.1.3	Cabling	All external cables entering the PDU shall pass through cable glands and ensure no cable entry opening shall be used without a cable gland.	Yes		
4.1.4	Cabling	Output connections shall be maintained as per the Station PDU schematic diagram 5 16 49 0671	Yes		
4.1.5	Cabling	Ensure lugs with sleeves / Ferrules are properly crimped and inserted into the terminal; no loose strands shall be left.	Ok		
4.1.6	Earthing	PDU units shall be connected to the ring earth/busbar using a 10 sq.mm green/yellow earthing wire, and bolts shall be tightened to a torque of 8 Nm with torque marking applied.	Ok		
4.1.7	Earthing	Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.	Ok		
4.1.8	Functionality	Functional testing shall be performed as per the PDU test procedure 5 53 69 0001.	Ok		
4.2 DC-DC Converter: Not Applicable					

5.0 Kavach Equipment

S.No	Aspect	Requirement	Status	Observation	Image
5.1 Kavach Unit					
5.1.1	Installation	Kavach unit shall be installed as per the approved floor plan drawing, mounted on the floor using M10 insulators, secured with M10 bolts, and tightened to a torque of 40 Nm as per diagram 5 16 76 0056.	Ok		
5.1.2	Cabling	All external cables shall enter through cable glands only. No unused cable entries left open. Ensure mill connector shall be locked properly.	Ok		

S.No	Aspect	Requirement	Status	Observation	Image
5.1.3	Earthing	Unit shall be connected to the ring earth/busbar using a 10 sq.mm green/yellow earthing wire, and bolts shall be tightened to a torque of 8 Nm with torque marking applied.	Ok		
5.1.4	Earthing	Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.	Ok		
5.1.5	Termination Unit	Station Kavach Termination Unit shall be wall-mounted near the Kavach unit using insulators, secured with M8 bolts, and tightened to a torque of 20 Nm as per diagram 5 16 76 0045.	Ok		
5.1.6	Termination Unit	OFC cables for SMOCIP and RTU shall be spliced as per diagram 5 16 49 0559, and proper bunching and routing shall be ensured.	Ok		
5.2 SMOCIP(Station Master's Operation-Cum-Indication Panel): Not Applicable					
5.3 GPS/GSM Antennas: Not Applicable					
5.4 RIU(Remote Interface Unit): Not Applicable					

6.0 RF Communication Equipment On Tower

S.No	Aspect	Requirement	Status	Observation	Image
6.1 RTU(Radio Tower Unit)					
6.1.1	RTU Fixing	Both RTUs shall be firmly secured to the tower platform using M12 bolts and nuts, and a torque of 85 Nm shall be applied as per diagram 5 16 67 0983.	Ok		
6.1.2	RTU Fixing	Ensure RTU doors are fully closed and locked.	Ok		
6.1.3	RTU Earthing	RTU shall be properly earthed by connecting a 35 sq.mm green to GI strip earthing conductor from the RTU earthing bolt to the designated earth pit-4, as per diagram 5 16 76 0043.	Ok		
6.1.4	RTU Earthing	Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.	Ok		

S.No	Aspect	Requirement	Status	Observation	Image
6.1.5	OFC cable termination	OFC cable from the Relay Room shall be spliced and terminated in the splice holder inside the RTU. (Ref. Drawing: 5 16 49 0559)	Ok		
6.1.6	OFC cable termination	OFC cables for RTU shall be spliced as per diagram 5 16 49 0559 and ensure bunching and routing shall be done properly.	Ok		
6.1.7	110V Power cable termination	Cable glands used for 110 V DC power cable entry into RTU shall be firmly tightened.	Ok		
6.1.8	110V Power cable termination	110 V DC power cables shall be terminated inside RTU as per approved drawing 5 16 49 0672.	Ok		
6.1.9	110V Power cable termination	Ensure lugs with sleeves / Ferrules are properly crimped and inserted into the terminal; no loose strands shall be left.	Ok		
6.1.10	Cabiling	LMR 600 connection with proper routing; and clamping; No Joints shall be done, as per Tower SOP 5 16 90 0018.	Ok		
6.2 RF Antenna Installation					
6.2.1	RF antenna installation and Audit	RF antenna installation and orientation shall be done as per 10.2dBi omni-directional anteena diagram 5 16 67 0983.	Ok		
6.2.2	RF antenna installation and Audit	RF antenna installation audit report from the installation contractor shall be available in WFMS and there shall be no open points in the audit report.	Ok		

7.0 OFC Networking Rack

S.No	Aspect	Requirement	Status	Observation	Image
7.1	Installation	Network rack shall be installed as per approved floor plan and installation shall be done as per diagram 5 16 76 0058.	Ok		
7.2	Installation	Patch cord routing shall be neat & bend radius to be maintained.	Ok		
7.3	Cabling	All external cables entering the network rack shall pass through the grommets.	Ok		
7.4	Cabling	OFC cables shall be marked using naming tie-tags for easy identification of default and standby links.	Ok		
7.5	Earthing	All networking modules inside the rack shall be connected to the rack chassis using 2.5 sq.mm green/yellow earthing wire.	Ok		
7.6	Earthing	Network rack shall be connected to ring earth using a 10 sq.mm green/yellow earthing wire.	Ok		
7.7	Earthing	Bolts shall be tightened to a torque of 8 Nm, and torque marking shall be applied using yellow paint.	Ok		
7.8	FDMS Box Installation in Network Rack	OFC cables shall be spliced in the FDMS box as per network drawing.	Ok		
7.9	FDMS Box Installation in Network Rack	FDMS boxes shall be clearly marked to identify up and down links.	Ok		
7.10	OFC cable continuity check, after splicing	OTDR test reports shall be available.	Ok		

8.0 Relay Rack (Relay Rack)	**Not Applicable**

9.0 Earthing

S.No	Aspect	Requirement	Status	Observation	Image
9.1	Installation	Earthing shall be done as per RDSO Spec RDSO/SPN/197/2008	Ok		
9.2	Earth Resistance Measurement	Earth resistance shall be measured using a calibrated earth resistance tester. The measured value shall be lessthan or equal to 1 Ohm.	Ok		
9.3	Test Reports	Earth resistance test reports shall be available	Ok		
9.4	Labeling	All earth pits, earthing conductors, and earth points shall be clearly labelled and identifiable.	Available		

10.0 Indoor Wiring / Cabling

S.No	Aspect	Requirement	Status	Observation	Image
10.1	Redundant cabling	Cabling shall be done as per station connectivity diagram 5 16 49 0614 Segregation of power & communication cables.	Ok		
10.2	SMOCIP	12 Core Signaling cable shall be used for button, counter & power supply. OFC armoured cables shall be used for communication.	Ok		
10.3	No Joints	No joints permitted except inside junction boxes or panels.	Ok		
10.4	Color Coding	Colour codes shall be followed for phase, neutral, and earth.	Ok		
10.5	Tagging/Labeling	Cables shall be tagged at both ends.	Ok		
10.6	Termination & C onnection	Proper lugs/ferrules shall be used and terminals tightened correctly.	Ok		

11.0 Outdoor Cabling

S.No	Aspect	Requirement	Status	Observation	Image
11.1	Cable route plan for OFC & Power cable (Relay Room to Tower)	Railway-approved outdoor signalling cable route plan shall be available.	Ok		
11.2	Cable route markers	Route markers shall be installed and clearly visible.	Ok		
11.3	Cable route markers	Where contractually required, underground RFID markers shall be installed and verified.	Ok		

12.0 RFID Tags

S.No	Aspect	Requirement	Status	Observation	Image
12.1	RFID Tag Installation	Ensure the main RFID tag is installed in the direction specified in the RFID tag layout, and a duplicate RFID tag shall be installed at a distance of 3–5 meters from the main tag, except for turnout tags.	Ok		
12.2	RFID Tag Installation	RFID tag mounting shall be firm and secure, ensuring no gap or mechanical play as per RFID tag installation procedure 5 53 76 0055.	Ok		
12.3	RFID Tag data Validation and Placement verification	System-generated RFID data validation report and Placement verification report shall be available.	Available		

13.0 Components Inspection

S.No	Aspect	Requirement	Status	Observation	Image
13.1	Materials Inspection (Which are not Inspected by RDSO or Consignee)	Verify that all subcontractors and HBL personnel use only approved make and part numbers as per the List of Acceptable I&C Materials EG-EN-FT-34.	Ok		

Note : This QC report is generated by the system and does not need a physical signature.